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"It all ends too soon" - Exploring stroke survivors and physiotherapists perspectives on stroke rehabilitation and the role of technology for promoting access to rehabilitation in the community

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Abstract

INTRODUCTION: Many stroke survivors do not receive the optimal levels of personalised therapy needed to support their recovery. Rehabilitation technology could offer a means to provide personalised guidance to stroke survivors struggling to access rehabilitation through other means.

AIMS: The aim of this study was to explore stroke survivor and therapist experiences of stroke rehabilitation post discharge into the community and their thoughts about the potential use of markerless, 3D motion capture technology which could support rehabilitation, to contribute towards a better understanding of how technology can support stroke rehabilitation.

METHODS: Focus groups and interviews were conducted with stroke survivors and therapists and the transcripts analysed using thematic analysis.

RESULTS: Six themes were identified across the data: access and continuity of care for stroke survivors in the community, stroke survivor confusion and lack of education impacting service navigation and ability to access rehabilitation services, stroke survivor access to non-NHS/private facilities, impact of Covid on stroke services, potential use of technology to support post-stroke rehabilitation in the community and, stroke survivors' willingness to try new technology.

CONCLUSIONS: Stroke survivors and therapists identified problems with stroke survivors accessing rehabilitation services but were positive about the potential for technology to support stroke rehabilitation.

Introduction

Stroke is a leading cause of long-term disability, often resulting in a combination of sensory-motor, communication, visual, and cognitive symptoms[1]. While some survivors recover, many remain with considerable levels of disability after stroke[1] and residual deficits that can deteriorate post-discharge from care[2]. These disabilities can impact on stroke survivors' quality of life and can limit their ability to reintegrate into society or return to employment. To maximise a patient's chances of recovering to their full potential, they need access to services which provide information and feedback about lifestyle, risk factor modification and therapy to address the impairments which are resultant from the stroke. A minimum dosage of therapy has been suggested, with a typical stroke patient recommended to receive 45 minutes of rehabilitation, five days a week[3]. However, this is rarely provided or made accessible for patients[4-6] and stroke survivors experience difficulties in adequately accessing rehabilitation services following discharge[4, 7-10]. Furthermore, time spent in therapy is not a sufficient metric for addressing impairments. Stroke survivors need time and correct information or feedback for their individual situation, otherwise there is a risk that they will not benefit.

Reduced access to effective rehabilitation services is likely to have long term effects on the stroke survivor as any gains can decay if access to services is restricted for a period of time. This can be further associated with poor recovery profiles and readmissions[6]. Services in the community can be difficult to access for discharged stroke survivors and in some cases are not present in local communities. Furthermore, patient perceptions of when they are ready to be discharged from the

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service may well be different from the threshold levels for end point of care. Removal of necessary support systems when they are still needed may well lead to deterioration and need for additional support further down the line.

Technology can be used to support rehabilitation for stroke survivors. Studies investigating the potential use of telerehabilitation or technology for stroke rehabilitation have been positive, see table 1.

Table 1 – Summary of the impact of technology in stroke rehabilitation

Benefit	Source References
Improvements to stroke survivor health, & wellbeing	11
Improved rehabilitation outcomes	12, 13
Greater autonomy and control for stroke survivors over their therapy	5, 6, 14-16
Provision of some structure and purpose to life following discharge	17
Technology increased long term access to services not otherwise available	11, 13, 18, 19
Improved the information provided to stroke survivors	18
Increased survivor sense of connection with the healthcare	6
Increased motivation and adherence	6, 15, 18
Secondary health benefits – early identification of other health conditions	11

Although these forms of technology support stroke survivors, they limit the stroke survivor to only being able to progress with knowledge from external therapists. Currently there isn't a facility that stroke survivors can use independently at home that offers a personalised service which supports the user with effective rehabilitation through an exercise programme, assessment of ability, monitor incremental improvement, and guide effective rehabilitation by providing feedback about knowledge or performance. Technology which contributes to knowledge of performance and result could support stroke survivors in carrying out meaningful independent rehabilitation and could supplement or support more traditional forms of therapy, increasing the amount of time available for stroke rehabilitation and leading to better outcome measures.

The VARE (Virtual Assessment and REhabilitation in Stroke) project is investigating the potential for interactive, personalised technology to support stroke survivors as they carry out rehabilitative exercises at home, alone or assisted by a carer. Using depth sensors for markerless, 3D motion capture, it may possible to accurately measure and assess user movements, provide personalised guidance and feedback in real time, and monitor incremental progression over time. Whilst not appropriate for all stroke survivors due to cognitive or severe physical disability, systems like this could benefit individuals who want additional support by providing individualised guidance and exercises. In addition to the physical benefits for stroke survivors, who could access therapy over a longer timeframe, this could also benefit both the healthcare services by reducing costs associated with inpatient care. The recent Covid-19 pandemic has significantly impacted stroke rehabilitation and its means of administration, restricting many in-person services and causing a surge in uptake of technology in all areas of healthcare[20], making it an opportune time to consider further integration of technology into rehabilitative therapy to provide care which has been further stretched due to Covid-19.

The aim of this study was to 1) better understand the experiences of stroke survivors and therapists with stroke rehabilitation in the community, 2) investigate the needs of stroke survivors who are currently unable to benefit from available technology and 3) to explore their thoughts about the

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potential use of markerless, 3D motion capture technology to support rehabilitation. This will contribute towards a better understanding of how technology could be developed to support stroke survivors and therapists in the future.

Methods

UK stroke survivors and therapists were invited to focus groups using mailing lists from either professional societies (ACPIN) or a stroke specialist therapy centre (ARNI). Participants were asked to talk about their experiences of post-stroke rehabilitation and their thoughts about potential technology to support them in this area. Participants were either 1) people with a previous stroke who had experienced rehabilitation in the community and who could clearly communicate about their experiences or 2) therapists with an interest in stroke.

Focus groups were made up of either physiotherapists or stroke survivors and included up to five participants at a time. The longest was an hour and a half. Focus groups were hosted virtually and were conducted by two members of the research team, with one clinical (AP), and one non-clinical (AFN) representative. Participants who couldn't attend the focus group were given the opportunity to participate via an interview. All encounters with participants were audio recorded, and anonymised transcripts were produced.

A thematic analysis was carried out by two members of the research team, one clinical (EL) and one non-clinical (AFN), following the guidelines laid out by Braun & Clarke[21]. Reflexive practices were built into the analysis process with both authors considering how their experiences and disciplinary knowledge affected their interpretation of the data. Codes were generated by the non-clinical author and were verified with the other authors. Themes and subthemes were generated iteratively by both researchers separately according to relevance to the research questions, and then were combined and refined by both researchers. Themes were further refined by all authors with regular revisiting of the data to confirm understanding and ensure that the analysis accurately represented participants' experiences.

Analysis

Participants

In total, ten stroke survivors took part in the stroke survivor focus groups (seven women, three men), and five therapists (all women, all physiotherapists working for the NHS in either acute neurorehabilitation roles or community stroke services, 3 specialist, 1 advanced) took part in the therapist focus groups. One participant participated through a phone interview, although this was not recorded, notes were taken during the interview to support analysis.

One participant with language impairments participated with their carer, who sometimes spoke on their behalf.

Findings

Six themes were identified across the stroke survivor and therapist focus groups:

- Access and continuity of care for stroke survivors in the community
- Stroke survivor confusion and lack of education impacting service navigation and ability to access rehabilitation services
- Stroke survivor access to non-NHS/private facilities
- Impact of Covid on stroke services
- Potential use of technology to support post-stroke rehabilitation in the community
- Stroke survivors' willingness to try new technology

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Access and continuity of care for stroke survivors in the community

All of the focus groups discussed the provision of services in the community, and community-based care was often compared to the facilities present in hospital or stroke-specific intermediate facility. Participants described positive experiences of stroke rehabilitation in the hospital, with regular, consistent therapy and rehabilitation.

"And I was in hospital for three months. During that time, the physiotherapy that I got was extremely good. I was in every day, Monday to Friday, and the weekends I did my own thing"

Stroke survivors described a big contrast between inpatient and community experiences, with many describing an inability to access sufficient care or rehabilitation after discharge. This was also raised by the therapists who discussed problems with transfer of care into the community. They described struggling to access follow up support when needed, particularly in areas where there were few community services, long waiting lists, strict entry criteria, or services being only available for fixed amounts of time with little to no flexibility to increase the amount of time accessed.

The stroke survivors described feeling that they were not given enough support post-discharge from hospital and were unable to access needed care or rehabilitation, feeling left on their own once they were discharged. This was most frequently characterised through: support being delayed due to long waiting lists, an inability to access community services appropriate for individual needs (either through lack of referral, services not being available in their area, or referral to services that were not specific to stroke survivor needs), or as services cut off after a fixed number of weeks without assessment of individual need.

"By the time I left hospital, I'd started walking with this good stick but when I got home the physical therapy went downhill. I was promised speech therapy but in the last three years I've had none."

"I think there's lots of people appreciate it all ends too soon. You know, I was on a quad stick and they were helping me to work to single point stick and then it was like, right we're gonna have to finish with you. I was like can you not carry on work with me 'til I can just use a single point stick right now? Their parting shot was 'you know what you need to do'"

Lack of access was identified across both local private and NHS services, although the latter was more frequently referenced by stroke survivors and therapists as it was their main provider of stroke rehabilitation. The consequences of this perceived lack of support were prevalent with nearly all of the stroke survivors describing physical impacts to their conditions associated with the reduction of rehabilitation, and consequent searching for further support. Stroke survivors' search for new services or information from other sources was often personally motivated and carried out online and through stroke support groups

"I just keep looking on the internet to see whether there's anything that's come along. Either technology-wise or any otherwise to see what improvements can be made or what research is going on and you know, if there's something that I think that might be useful to me, then I'll join in with it."

Most stroke survivors mentioned wanting to do more to support themselves with their post-stroke rehabilitation but described limited access to community services due to lack of available facilities and difficulty travelling to the location. Participants described searching for stroke specific programmes, community programmes, alternative therapists/trainers specialised in stroke, or research projects that they could be part of to receive additional support or information about their conditions. These individuals expressed high levels of determination to support and rehabilitate

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themselves as much as possible, despite many reporting that their therapists had said that they would not be able to progress further.

"I think I find it interesting that everybody in this group has said that they want to progress more. They want to do more but there aren't the facilities."

"And after observing [Participant], the physio said there's nothing he can do. OK, so he discharged him […] [Participant] is mobile now […] that's all [Participant]'s doing."

They described their recovery journey in individual terms, emphasising the "hard work" that they had put in to support themselves either as personal practice or through support from non-therapist figures they had gone to see. Individuals also described the physical benefit of this individual work and were proud of the improvements which they saw as a result.

The search for beneficial support or further information about how to improve personal condition was mentioned by every stroke survivor, but this individual work was guided by financial resources. Those who were financially able described attending local services, stroke specific facilities, or buying technology to support their development at home. However, many of the stroke survivors described limitations in the resources they could access due to cost. Many engaged in research projects to access therapy that they would be unable to afford otherwise with one participant explicitly mentioning four research projects that they had been part of (not including this study) and indicating that they had also participated in others, describing how these projects allowed them to engage with therapy that they would not have been able to otherwise access and the perceived benefits which came from the additional support.

Stroke survivor confusion and lack of education impacting service navigation and ability to access rehabilitation services

Despite individually searching for further services and resources, many stroke survivors described a lack of knowledge about their condition and how to navigate the healthcare system to access services. They expressed how they had experienced feelings of uncertainty and confusion due to a lack of accessible knowledge about their stroke, treatment options open to them, how their prescribed exercises were related to their stroke, and how these things might change in the future. This lack of education was attributed to staff not explaining their situation to them thoroughly:

*"Nobody ever really kind of explained that all to me and the purpose of these exercises. They'd say to this, do that, but nobody actually explained **why** we were doing it."*

"I think it's changing now, but in hospital, they'd sort of tell you reached a plateau and what you didn't do in the first six months or so, you would never be able to do. […] So sometimes you come out of hospital with this belief that, or you finish your community physio with the belief that that's it. You're done. Now you won't go any further."

An element of this was suggested to be due to the clinicians' language which some stroke survivors described as complex and difficult to understand. They perceived the knowledge about their condition to be inaccessible, leaving them feeling ostracised and unable to access the support and information they needed particularly at the point of discharge from hospital.

"I don't have much movement in my arm, but nobody's really explained to me if there's anything I can do about it. I just wondered if it would be, sort of like this, just dangling down my side all my life, I don't know, and when you ask questions, they say 'well try an orthotic of some sort'."

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A particular area of confusion for many pertained to the care pathways within the NHS and how to navigate the community services. Several participants discussed not understanding referrals or knowing how to access further services when needed, with no signposting or further information from therapists. They described feeling abandoned and frustrated because they felt that they had to manage the process alone and at a disadvantage because they were unaware of the processes to go through or whether they could ask for further help.

"[The NHS] don't actually supply you with any information at all when you're discharged. There ought to be some sort of support chain once you're actually discharged, even if they just give you a few names and addresses, but there's nothing."

One individual described problems with their therapist and needing further support but being unaware of how to access it themselves. While they did eventually receive a referral to a stroke specialist clinic, they had been given the referral by a family member, not their GP, and so described their process as "a bit cheating" because they perceived that it would have been inaccessible without the family member's intervention.

Stroke survivor access to non-NHS/private facilities

Private services were mentioned by both stroke survivors and therapists as an alternative option for rehabilitation post-discharge. While both acknowledged the presence of these alternatives, and many stroke survivors suggested that some private services were immensely beneficial for their development, there was a lot of discussion about whether these facilities were accessible to stroke survivors.

Therapists said that they were aware of private options for stroke survivors but were not sure whether patients were able to access these services. Similarly, they were not sure if stroke survivors would have the equipment or skills to access digital or online services. Stroke survivors confirmed this in their focus groups. Private services were mentioned in all focus groups and those who used them mentioned benefitting from them. However, the groups described multiple factors which limited access to private services including: unequal distribution of services across the country, limited places on services, personal problems accessing services, and cost. Cost was the most mentioned barrier to accessing any stroke services. Several individuals described taking part in research projects specifically to access care that would otherwise be financially inaccessible, while others described using carers or more financially accessible support like gym instructors to access exercise prescription and support because they couldn't afford the expense of private physiotherapy.

Impact of Covid-19 on stroke services

Both groups of participants discussed how the Covid-19 pandemic had limited access or provision of stroke services in the UK and made navigating the systems more difficult. Several stroke survivors said that the pandemic stopped therapists or support workers visiting their houses and noted that their physical progress had suffered because they hadn't been able to have people working with them.

"I found that I cannot do exercises on my own but if I have somebody there to help me, it's spurs me on to do more […] but then when the pandemic started, they said they couldn't do it again…"

They described how private programmes had stopped or had introduced entry restrictions to comply with social distancing requirements which limited the amount of supported rehabilitation that was available. Therapists also mentioned that community services had been restricted, specifically

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mentioning the national exercise referral scheme, and described how this led to increases in the amount of time that stroke survivors spent without access to therapist-led programmes.

Therapists described how the pandemic had altered priorities in hospitals and how it had affected stroke survivor care. They said that the Covid pandemic had shifted the focus from rehabilitation to progressing patients so that they could be moved into community where they would be at lower risk of catching the virus.

"Our main focus was shifted from rehab to just getting people out of the door"

This shift in priority impacted stroke therapy and reduced the amount of time stroke survivors received with therapists. The therapists also discussed how the Covid restrictions meant that relatives could not visit stroke survivors in hospital which had an emotional impact on the families and made family/carer education difficult impacting patients' transition into the home.

While all participants agreed that Covid had been difficult, there were some positive elements attributed to it. Many stroke survivors said that the social distancing regulations had caused services to introduce virtual methods of provision and explained how this was beneficial for them as it allowed them to access services that they would not have been able to attend in person.

"[…] Since Covid, obviously, that hasn't been able to happen, but I found lots of things online, so I actually think it's been a positive thing that people, organizations have been forced to use technology to deliver their services. My physio is based down near Northampton. I've never met them; it's all done online. That wouldn't have happened for me because where I am up in the North-West, there just isn't the services available."

Potential use of technology to support post-stroke rehabilitation in the community

All participants were positive about the prospect of technology which could be used to support post-stroke rehabilitation in the community. Both stroke survivors and therapists thought that a telehealth-based system which could provide visual guidance and feedback for the stroke survivor, has the potential to be useful and beneficial for stroke survivors. In particular they were positive about being able to continue to receive personalised guidance and feedback about quality of movement when unable to attend or see physical therapists.

"So I'm a great believer in technology and making technology work and I think for me I've made it work for me. Now organizations that are embracing technology in, according to environment, that's a really good thing and I hope it continues."

Participating therapists indicated that they thought that integration of technology into stroke rehabilitation was expected, a natural progression of healthcare, and could be ideal for a patient who needs to be discharged from a service but who would benefit from further therapy and who could manage exercises by themselves. They suggested that technology supporting data collection of assessment of survivors' actions as they carried out exercises would be a beneficial attribute for both therapist and stroke survivor and would allow both to monitor progression over time. Having a record of development was raised as a potential motivator for stroke survivors and was thought likely to increase adherence.

Stroke survivors suggested that technology would empower the users by educating them and their carers about the rehabilitation process so that they were better able to support themselves without needing a therapist to be physically present. They talked positively about the potential of being

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supported to work independently and self-manage their rehabilitation rather than being a passive receiver limited to improvement only when they could access therapy through stroke services.

Both stroke survivors and therapists had reservations about the inclusion of technology in post-stroke rehabilitation. Stroke survivors were concerned that technology would not provide the same level of tailored support that an in-person therapist would administer. This concern could be alleviated, they suggested, if the technology was monitored by a therapist. Therapists suggested that technology may not be accessible to all patients and that some survivors may have concerns about engaging with technology or may not have the equipment or skills to access technology-based therapy. They suggested that in general a younger cohort might be better able to use technology than an older cohort, and that it might be better for higher functioning patients and prove difficult for patients with cognitive issues or perceptual problems. Users would also have to be self-motivated to engage with it because there may not be another driving force to encourage them to interact with it. Stroke survivors also discussed the holistic elements of therapy, saying that also act as a point of social contact for individuals who can be lonely following discharge. Although the technology could empower survivors to control their own rehabilitation, many were concerned that use of the technology could result in emotional challenges due to fewer social interactions. Cost was raised as a significant barrier for uptake by both stroke survivors and therapists.

Stroke survivor willingness to try new technology

Despite their reservations, all stroke survivors suggested that they would be likely to try new stroke rehabilitation technology. However, it was apparent that this was associated with an inability to access other services.

"If we have the app when we leave hospital, […] he can be engaged with physio straightaway rather than being left in fact for 2-3 years with nothing."

The stroke survivors were willing to try anything which had the potential to support them with their rehabilitation and appreciated technology which they could use independently as it would ensure that they had continuous, stroke-specific rehabilitation support post-discharge either while on waiting lists for services or otherwise unable to access support.

Discussion

This study aimed to document stroke survivors' and therapists' perceptions of stroke rehabilitation, with a view to better understanding whether markerless, 3D motion capture technology could support stroke rehabilitation at home. Six themes were developed from the data which identified difficulties that stroke survivors face accessing stroke rehabilitation and their thoughts about the prospect of incorporating technology into rehabilitative practice in the community.

Developments in stroke services and interventions can benefit stroke survivors if they are able to access them at the right time, but they are not ubiquitous across the care pathway. The focus groups drew attention to the inability of stroke survivors to access sufficient rehabilitation and the consequent physical impact. Many participants described struggling to access further support when needed, either from limited services available in their area or due to difficulties navigating the healthcare system. While not explored in this study, this may well be worse for those with higher levels of disability or who have lower levels of support around them. Although, the participants here may be those who are more likely to engage with rehabilitation in general, the discussion highlighted that there are groups of stroke survivors who want to improve but who are not able to access facilities to support them in rehabilitation, a finding seen in other studies[5] and highlighted how little has changed in stroke rehabilitation services since the National Stroke strategy was published in

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2007[22]. The stroke survivors in this study expressed a desire to improve and were willing to take personal responsibility for their rehabilitation but were unable to access appropriate support through their current pathways, which were predominantly NHS services. For those without additional funds or knowledge to access private services, research studies were the only way they could access guidance (and many benefited from this engagement). While not all survivors want additional therapy, it was clear in the focus groups that many stroke survivors felt they would benefit from further rehabilitation with integrated feedback, and that there are a number of stroke survivors who would like to be able to access this kind of rehabilitative support. For many of these individuals, technology could offer a good way to access additional, personal rehabilitation when they are ready to engage with it, even when discharged.

A common discussion point raised by stroke survivors was a lack of knowledge about their condition, how to manage it, and how to access services and navigate the healthcare system. Stroke survivors often described not feeling like they received enough information about their situation and recounted problems accessing suitable services, suggesting that they felt cut off by not being able to navigate the healthcare system. Interestingly, those who had managed to access services alluded to feelings of guilt, perceiving themselves as 'cheating' when they accessed support through family or friends rather than through standard routes. A perceived lack of knowledge about strokes, and a desire for better provision of knowledge about the condition to be provided to stroke survivors upon discharge has been identified in earlier research[23, 24]. This study highlights the need for clinics to provide detailed, accessible knowledge to stroke survivors and to ensure that healthcare professionals have up-to-date knowledge about strokes and stroke services so that they are able to make accurate knowledge accessible to the patients and families they work with. It is likely that, the chronological point at which information is provided to patients and family/carers is equally important, given the readiness of patients to make sense of the information during the initial phases after the events of a stroke when they may have more pressing priorities at hand. The disconnection from healthcare services, driven by a lack of knowledge about access and navigation, is a novel finding which should be viewed with concern, as it could have more serious health connotations for individuals who may be missing out on necessary support. Markerless 3D motion capture technology could also potentially act as a point of later contact, ensuring that stroke survivors and their families are able to access information about rehabilitation and wider support services when they are ready to receive and engage with it, an unintended potential benefit. The ability to monitor incremental improvement could also support stroke survivors in tracking their developments and sharing their progress with others, and in identifying any problems which occur over time. Historic data could also support stroke survivors if they seek additional support, offering a quantitative assessment of their capabilities which can be used by therapists to assess their suitability against service thresholds.

It is important that technology is accessible, easy to use, and meets user needs. Cognitive ability and capacity will influence the degree that any technology can be engaged with, but there are a wide range of factors that should also be considered as part of the development of the device. Of these, cost was regularly mentioned by stroke survivors and therapists in the focus groups, including cost of the technology, the device to host it, and the resources and internet infrastructure to support it, similarly identified in earlier studies [10, 25]. Other elements that should be considered as part of the development of rehabilitation technology include the role that rehabilitation plays for stroke survivors and whether survivors will be safe to use the technology alone. In the focus groups stroke survivors suggested that therapy provided a social link beyond the health benefits, particularly where survivors were unable to access former socialising opportunities. This has been documented before[10], and while socialising is not the primary function of rehabilitation, the impact of loss of social links can impact on survivors' wellbeing and so should continue to be considered while

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developing stroke services. Safety should also be considered and incorporated to ensure survivors are not at risk while exercising[14]. Current rehabilitation exercises consider what the stroke survivor will be able to do alone with minimal risk of harm and these thoughts should be integrated into technology, considering risk of falls and whether the survivor is alone or accompanied by a carer when presenting exercises to the user. Other elements to be considered include the need for the service to last beyond the usable life of the technological device it is hosted on. Technology is developing at an intense rate, so it is important to ensure that any system built for stroke rehabilitation is integrative with other services and can remain useful or not made redundant as technology develops. A model with integrated updates for developing hardware will ensure that the technology remains relevant and useful for stroke survivors into the future. Wherever possible, effort should be made to utilise the most affordable and accessible materials possible to minimise potential users from being ostracised from healthcare services.

All participants were positive about the prospect of integrating technology into stroke rehabilitation in the community, particularly technology which could provide guidance and personalised feedback. Everyone expressed interest in trialling future technology as part of the project. This was largely expected given positive results of previous research [10, 16, 26] and the increasing use of technology to administer healthcare services in recent years. However, it was encouraging to see a high level of enthusiasm within this study. Stroke survivors suggested that the provision of this technology would empower them to take control of their therapy and be able to support themselves in situations where they were unable to access in-person therapy, and this may positively impact on adherence and could have wider impacts on stroke survivor health and wellbeing, and potentially reduce their need to access other health services. However, it should be acknowledged that the participant base which we were working with during this study were individuals who were engaged in stroke rehabilitation research and already had technology that they were comfortable using. Therefore, these results only reflect the thoughts of motivated participants. Additional work needs to be carried out to identify the thoughts and experiences of stroke survivors and stroke therapists who may be less motivated or who are less able to engage with technology.

From these focus groups, it is apparent that there is a need for greater support with stroke rehabilitation in the community and markerless 3D motion capture technology could be a tool which could support stroke survivors and therapists in these areas. Accessibly priced technology could provide a means to provide continuous, stroke-specific therapy as stroke survivors leave hospital, and the findings of this study suggest that both stroke survivors and therapists would be willing to trial new technology to access this support. The ability to personalise a programme and adapt it to the users' personal developments could support and empower stroke survivors with their rehabilitation and help maintain access to therapy even when services are limited or inaccessible. Incremental change, and the ability to observe this in a comprehensible format, was suggested to be important to the stroke survivors in the focus groups, and likely to contribute towards increased adherence over time. Markerless 3D motion capture technology could support the stroke survivors with this element of rehabilitation and thus encourage the maintenance of rehabilitation practice over time. This could contribute to reducing deterioration caused by lack of access to services and would also ensure that patients and their family had access to support and information when they were ready to engage with it. This education is important because stroke survivor needs will change throughout their life[23], and will support them in managing their condition and know the best services to support themselves if needed.

Limitations

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The study was impacted by Covid-19 restrictions and this significantly affected our ability to recruit. While we received some interest from stroke survivors, recruitment from therapists was limited and not very diverse. It would have been beneficial to include a wider range of therapists, particularly those from different backgrounds, to gain additional insight into their understanding of post-stroke rehabilitation.

The virtual participation was beneficial for recruiting stroke survivors as it allowed us to talk to individuals across the country, some of whom said they would have been unable to participate had we conducted the focus groups in person. However, we acknowledge that there was a sampling bias in favour of stroke survivors and therapists who were already comfortable with the use of technology (and who have the finances and resources accessible to engage with technology in their homes), and stroke survivors who are motivated to engage with rehabilitation and technology-based exercises. Future studies should investigate populations who may be less willing to engage with rehabilitation or technology to determine factors which would affect their likelihood to engage with technological methods of therapy.

Conclusion

This paper investigated the perspectives of stroke survivors and therapists with community stroke rehabilitation experience and their thoughts about the potential benefit of markerless, 3D motion capture technology for their rehabilitation exercises following discharge from acute care. Stroke survivors and therapists discussed problems with stroke survivors accessing services but were positive about the prospect of incorporating technology into their future, particularly where it could supplement services that were inaccessible either due to cost, availability or location. This investigation also identified changes to stroke rehabilitation and the delivery of therapy which have been brought about by the Covid-19 pandemic, noting the widespread uptake of video calls and telehealth-based methods to engage with stroke rehabilitation. As the country moves out from the pandemic, further development and integration of technology and telerehabilitation to support stroke rehabilitation services could be a solution to empower stroke survivors in their rehabilitation and ensure that they have a means to access rehabilitation continuously from discharge from hospital. For therapists, technology provides an additional tool that can be used to support stroke survivors in the community. While care needs to be taken to ensure that the technology is accessible to all users, this is an area to be expanded upon in the future.

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References

1. Lawrence ES, Coshall C, Dundas R, et al. Estimates of the prevalence of acute stroke impairments and disability in a multiethnic population. *Stroke* 2001; 32: 1279-1284.
2. Van De Port IG, Kwakkel G, Van Wijk I, et al. Susceptibility to deterioration of mobility long-term after stroke: a prospective cohort study. *Stroke* 2006; 37: 167-171.
3. Stroke II, Bowen A, James M, et al. National Clinical Guideline for Stroke. In: 2016, Royal College of Physicians.
4. *Sentinel Stroke National Audit Programme (SSNAP). Clinical audit Aut-Nov 2016 (Public Report)*. 2016.

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It is made available under a CC-BY-ND 4.0 International license .

5. Marwaa MN, Ytterberg C and Guidetti S. Significant others' perspectives on person-centred information and communication technology in stroke rehabilitation–a grounded theory study. *Disability and rehabilitation* 2020; 42: 2115-2122.
6. Marwaa MN, Kristensen HK, Guidetti S, et al. Physiotherapists' and occupational therapists' perspectives on information and communication technology in stroke rehabilitation. *Plos one* 2020; 15: e0236831.
7. Hayward KS and Brauer SG. Dose of arm activity training during acute and subacute rehabilitation post stroke: a systematic review of the literature. *Clinical rehabilitation* 2015; 29: 1234-1243.
8. Stockley R, Peel R, Jarvis K, et al. Current therapy for the upper limb after stroke: a cross-sectional survey of UK therapists. *BMJ open* 2019; 9: e030262.
9. Holden MK, Dyar TA and Dayan-Cimadoro L. Telerehabilitation using a virtual environment improves upper extremity function in patients with stroke. *IEEE transactions on neural systems and rehabilitation engineering* 2007; 15: 36-42.
10. Edgar MC, Monsees S, Rhebergen J, et al. Telerehabilitation in stroke recovery: a survey on access and willingness to use low-cost consumer technologies. *Telemedicine and e-Health* 2017; 23: 421-429.
11. Johansson T and Wild C. Telerehabilitation in stroke care–a systematic review. *Journal of telemedicine and telecare* 2011; 17: 1-6.
12. Ameer K and Ali K. iPad use in stroke neuro-rehabilitation. *Geriatrics* 2017; 2: 2.
13. Crotty M, Killington M, van den Berg M, et al. Telerehabilitation for older people using off-the-shelf applications: acceptability and feasibility. *Journal of telemedicine and telecare* 2014; 20: 370-376.
14. Luker J, Lynch E, Bernhardsson S, et al. Stroke survivors' experiences of physical rehabilitation: a systematic review of qualitative studies. *Archives of physical medicine and rehabilitation* 2015; 96: 1698-1708. e1610.
15. Mallet KH, Shamloul RM, Corbett D, et al. Recovernow: Feasibility of a mobile tablet-based rehabilitation intervention to treat post-stroke communication deficits in the acute care setting. *PloS one* 2016; 11: e0167950.
16. Pugliese M, Ramsay T, Johnson D, et al. Mobile tablet-based therapies following stroke: A systematic scoping review of administrative methods and patient experiences. *PloS one* 2018; 13: e0191566.
17. Damush TM, Plue L, Bakas T, et al. Barriers and facilitators to exercise among stroke survivors. *Rehabilitation nursing* 2007; 32: 253-262.
18. Brouns B, Meesters JJ, Wentink MM, et al. Why the uptake of eRehabilitation programs in stroke care is so difficult—a focus group study in the Netherlands. *Implementation Science* 2018; 13: 1-11.
19. Pugliese M, Ramsay T, Shamloul R, et al. RecoverNow: A mobile tablet-based therapy platform for early stroke rehabilitation. *PloS one* 2019; 14: e0210725.
20. Bhaskar S, Bradley S, Chattu VK, et al. Telemedicine across the globe-position paper from the COVID-19 pandemic health system resilience PROGRAM (REPROGRAM) international consortium (Part 1). *Frontiers in public health* 2020; 8: 644.
21. Braun V and Clarke V. *Thematic analysis: A practical guide*. Sage, 2021.
22. Health Do. *National Stroke Strategy*. 2007.
23. Eames S, Hoffmann T, Worrall L, et al. Stroke patients' and carers' perception of barriers to accessing stroke information. *Topics in stroke rehabilitation* 2010; 17: 69-78.
24. Rodgers H, Bond S and Curless R. Inadequacies in the provision of information to stroke patients and their families. *Age and ageing* 2001; 30: 129-133.
25. Chen CC and Bode RK. Factors influencing therapists' decision-making in the acceptance of new technology devices in stroke rehabilitation. *American journal of physical medicine & rehabilitation* 2011; 90: 415-425.

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It is made available under a CC-BY-ND 4.0 International license .

26. Mallet K, Shamloul R, Pugliese M, et al. RecoverNow: A patient perspective on the delivery of mobile tablet-based stroke rehabilitation in the acute care setting. *International Journal of Stroke* 2019; 14: 174-179.